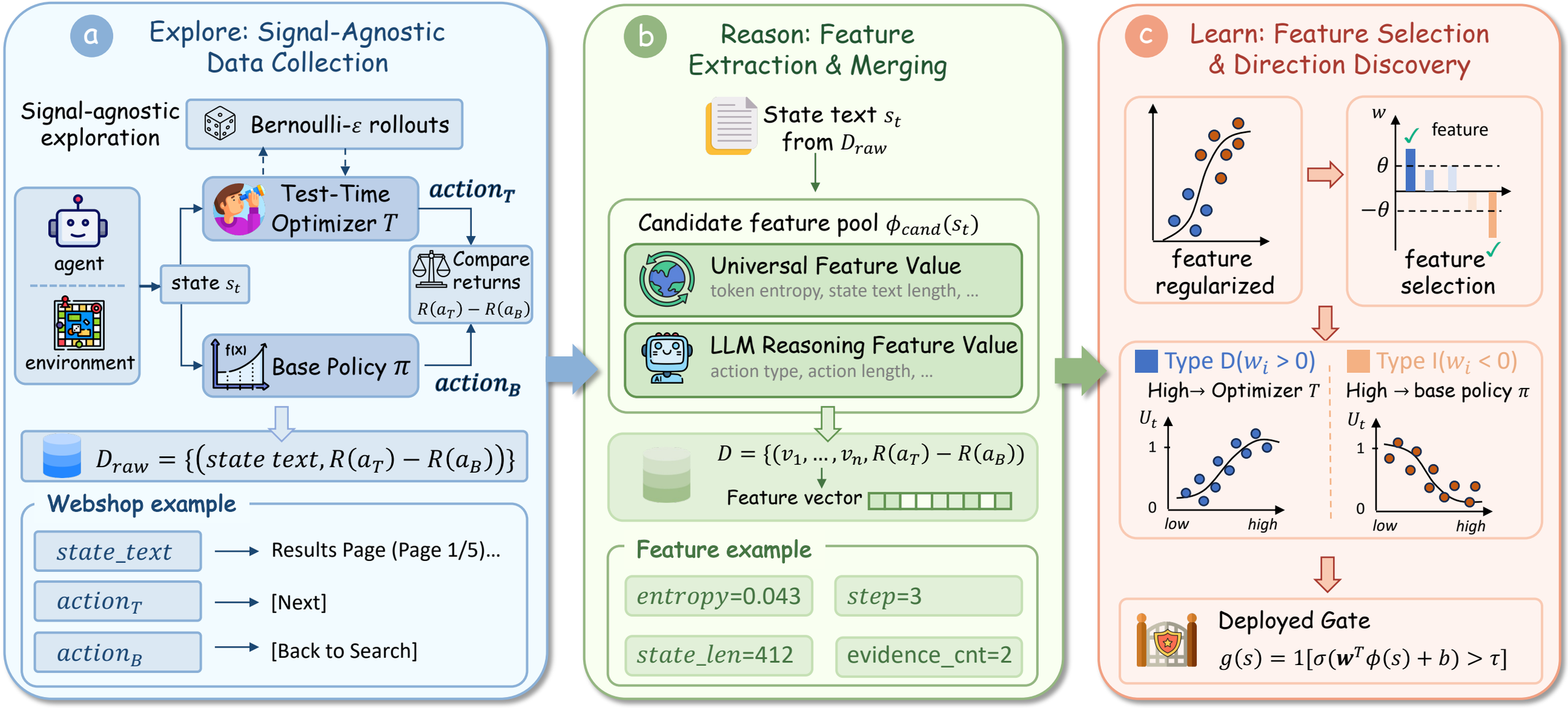
a
Explore: Signal-Agnostic Data Collection
Signal-agnostic exploration
Bernoulli-ε rollouts
Test-Time Optimizer T
action_T
agent
environment
state s_t
Compare returns
R(a_T) − R(a_B)
f(x)
Base Policy π
action_B
D_raw = {(state text, R(a_T) − R(a_B))}
Webshop example
state_text → Results Page (Page 1/5)...
action_T → [Next]
action_B → [Back to Search]
b
Reason: Feature Extraction & Merging
State text s_t from D_raw
Candidate feature pool φ_cand(s_t)
Universal Feature Value
token entropy, state text length, ...
LLM Reasoning Feature Value
action type, action length, ...
D = {(v_1, ..., v_n, R(a_T) − R(a_B))}
Feature vector
Feature example
entropy=0.043
step=3
state_len=412
evidence_cnt=2
c
Learn: Feature Selection & Direction Discovery
feature regularized
w
feature
θ
−θ
feature selection
Type D(w_i > 0)
High→ Optimizer T
Type I(w_i < 0)
High → base policy π
U_t
1
0
low
high
Deployed Gate
g(s) = 1[σ(w^T φ(s) + b) > τ]